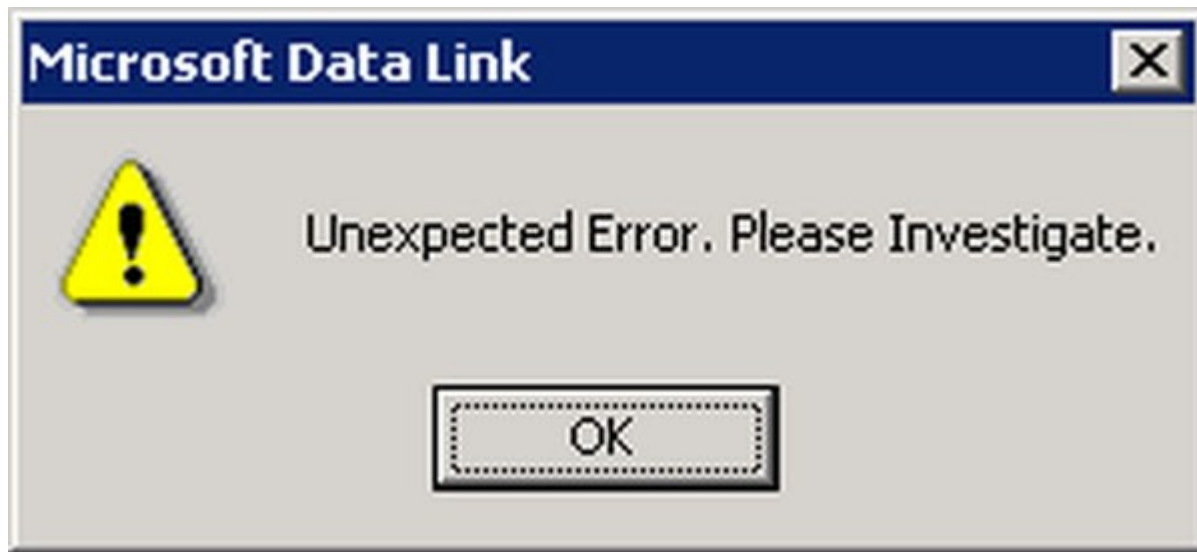


Experimental Error

How *good* is my data?



Experimental Error

Why do you repeat your experiment?

How come when you do an experiment you do not get the exact same result every time?

Where does error come from?

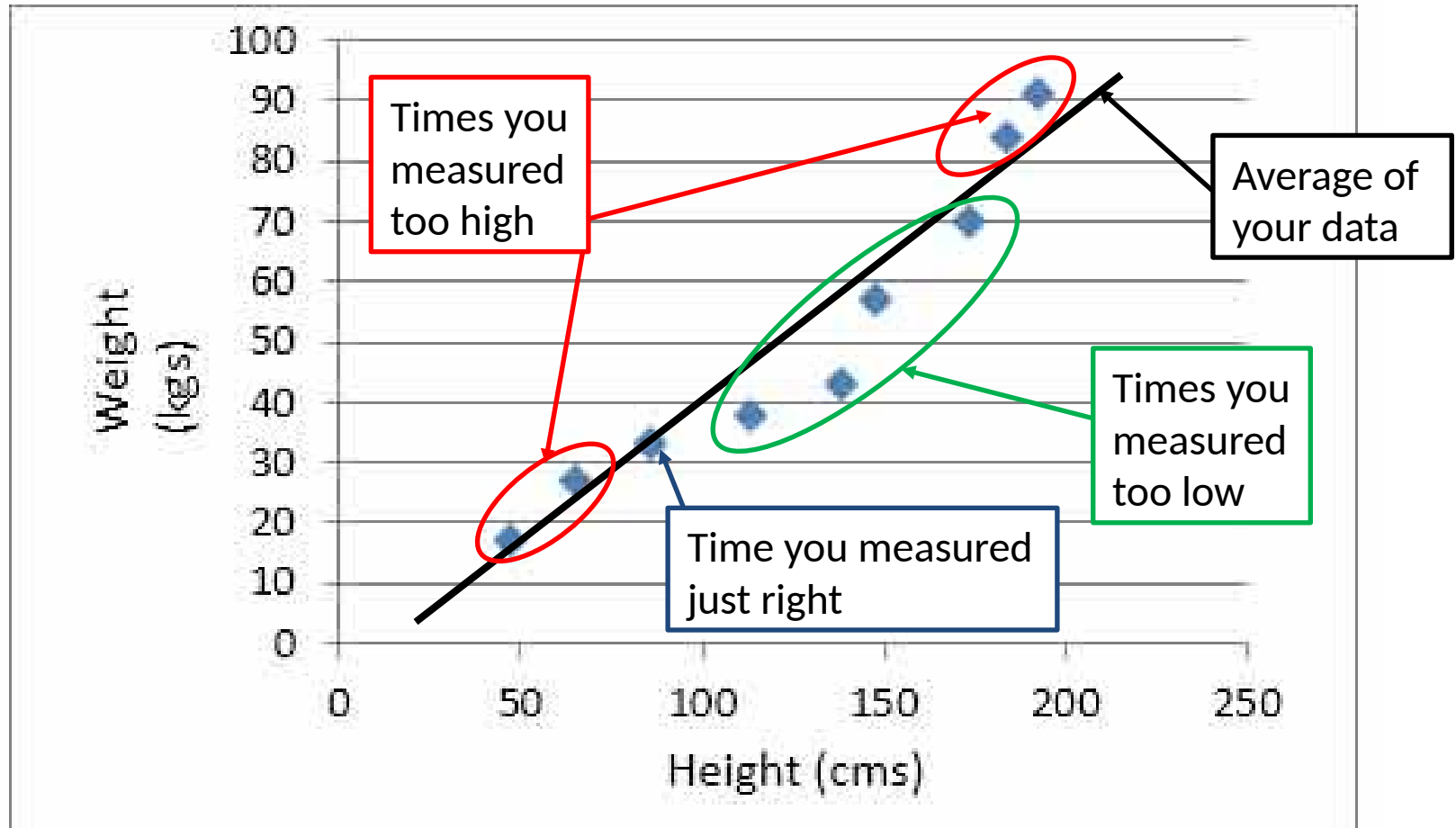
Repeating Experiments

Allows for statistics to get you closer to the correct answer

When you do multiple experiments you might be higher than actual and lower than actual approximately the same amount of times

An average of your results gets you close to the true value (Graphically this is a trend line!)

Trendline is an Average



Sources of Error

You (reaction time, sight, etc.)

Tools you use (limitations come from how small of measurements they can take)

Manipulation/changing the environment slightly
(moving something from its original position)

External influences

Experimental design (not well controlled)

Analyzing Data

Look for:

Precision

Accuracy

Error

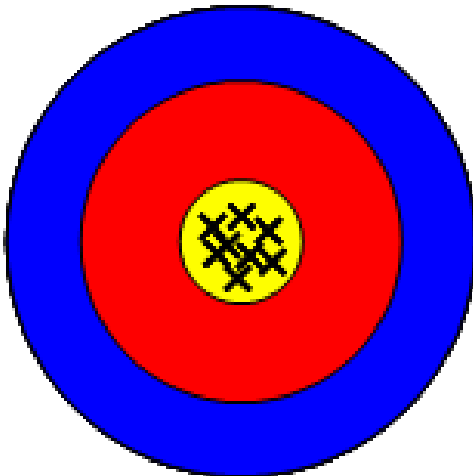
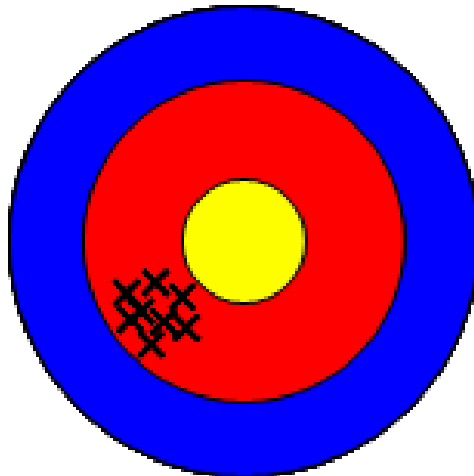
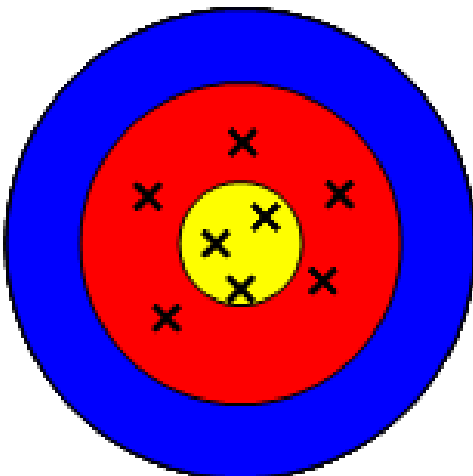
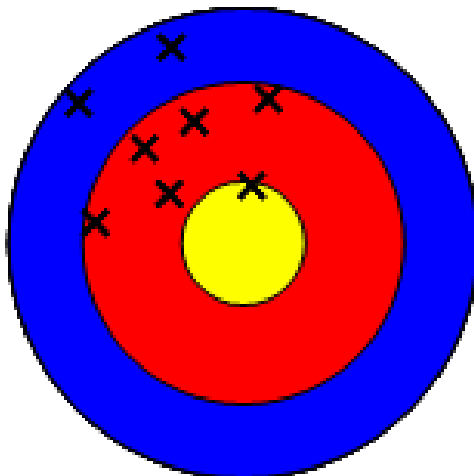
% Error



Analyzing Data

Precision: how close your data is when you repeat the experiment

Accuracy: how close your data is to the expected value

	Accurate	Inaccurate (systematic error)
Precise		
Imprecise (reproducibility error)		

Precision – During Experiment

Repeatability

Can be observed and potentially corrected during experiment

If you notice an “outlier” or a data point that is totally off, just repeat measurement

Trial #	Time to finish line (s)
1	12
2	27
3	13
4	11

Precision – During Experiment

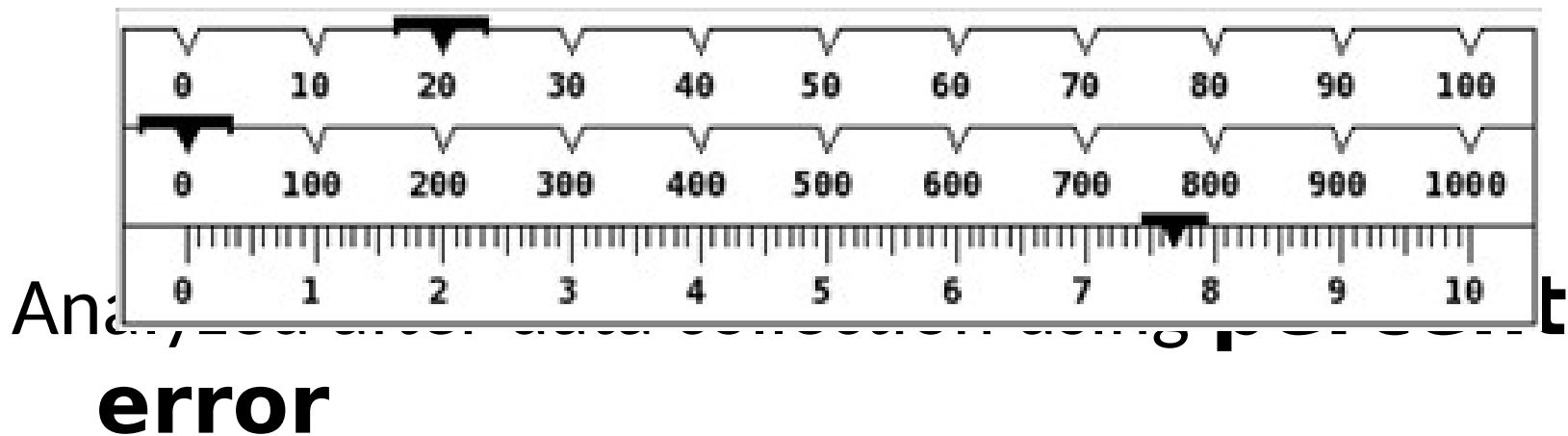
When measuring the room's volume with meter sticks, which method do you think will give you a more precise measurement?

Placing the meter sticks end to end while holding them in the air

Placing the meter sticks end to end while laying them on the floor

Accuracy

Proximity (closeness) to target





Error and Percent Error

$$\text{error} = |\text{measured} - \text{actual}|$$

Absolute
value means
NO
NEGATIVES

Why can't we
just use this
value?

$$\% \text{ error} = |\text{measured} - \text{actual}| \div \text{actual} \times 100$$

No units,
just put a
% at the
end

Why is “Percent Error” Important?

A blue whale has a mass of 180,000 Kg. YOU measured it to be 175,000 Kg

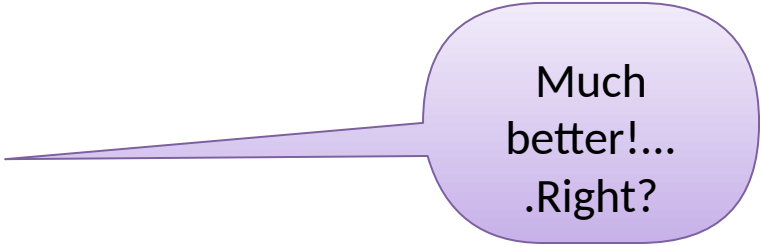
What is your error?

$$180,000 \text{ kg} - 175,000 \text{ kg} = 5,000 \text{ kg}$$

Your cat has a mass of 8 Kg. YOU measured it to be 7 Kg.

What is your error?

$$8 \text{ kg} - 7 \text{ kg} = 1 \text{ kg}$$



Much
better!...
.Right?

Accuracy and Percent Error

A blue whale has a mass of 180,000 Kg. YOU measured it to be 175,000 Kg

What is your error?

5,000

What is your % error?

Your cat has a mass of 8 Kg. YOU measured it to be 7 Kg.

What is your error?

1

What is your % error?

THE CLOSER YOUR PERCENT ERROR IS TO 0, THE MORE ACCURATE YOUR DATA IS!